

AS3935 Lightning Detector

Energy Values Explained

What is Lightning Energy?

The **energy** (also called **intensity**) value from the AS3935 lightning detector is not a physical measurement in Joules or Watts. It's a **relative dimensionless scale** that represents the **strength of the electromagnetic signal** received by the detector's antenna.

Detection Mechanism

The AS3935 chip detects lightning by sensing electromagnetic waves (radio frequency noise) produced by lightning discharges:

1. Lightning Strike
2. Electromagnetic Wave (RF energy)
3. AS3935 Antenna receives signal
4. Internal algorithm processes signal pattern
5. Outputs: Distance estimate + Energy value

Energy Scale - 21-bit Resolution

The AS3935 outputs a **21-bit energy value** ranging from **0 to 2,097,151**

21 bits = 2^{21} = 2,097,152 possible values (0 to 2,097,151)

What Influences Energy Values

The energy reading is affected by:

Strike distance Closer = Higher energy (inverse square law)

Strike power More powerful discharge = Higher energy

Strike type Cloud-to-ground > cloud-to-cloud (typically)

Antenna sensitivity Higher gain settings = Higher readings

Environmental noise Can inflate or deflate readings

Antenna tuning Proper tuning = More accurate relative values

Important Limitations

The energy value is NOT calibrated to absolute units

You cannot convert it to Joules, Watts, or Amperes

You cannot compare values between different AS3935 installations (antenna variations affect readings)

It's only useful for relative comparisons within the same installation

The manufacturer (AMS/ScioSense) does not publish the conversion formula to absolute energy

AS3935 Internal Processing

The chip performs sophisticated signal analysis:

1. **Pattern Recognition:** Distinguishes lightning from noise (motors, appliances, RF interference)
2. **Energy Calculation:** Measures peak amplitude of the received RF waveform
3. **False Positive Rejection:** Filters out man-made disturbances
4. **Statistical Averaging:** Uses recent strikes (last 15 seconds) to improve distance estimates

Practical Use Cases

Storm Tracking

Watch energy values increase then decrease to track storm movement:

Rising energy + decreasing distance = Storm approaching

High energy + stable short distance = Storm overhead

Falling energy + increasing distance = Storm receding

Fire Danger Modeling

Higher energy strikes are more likely to start fires:

Dry fuel + high energy strikes (>1,000,000 or >50% max) = Elevated fire risk

Used in Stratus Fire Danger Index calculation